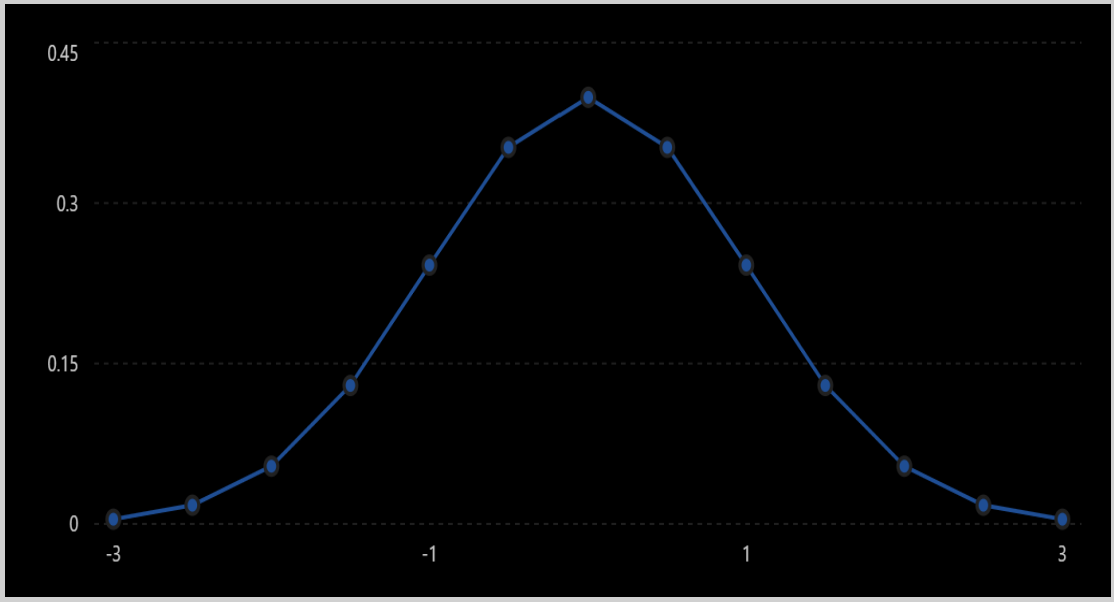
Normal Distribution, Standard Normal Distribution its Properties

Normal Distribution in Depth

A normal Distribution is continuous probability distribution with a symmetric, bell-shaped curve.

Technically:

It is a contnuous probability distribution characterised by a symmetrical bell shaped probability density function determined by it mean (μ) and standard deviation (σ)



"The curve is highest in the center and decreases as we move away from the center"

***** Properties of Normal Distribution**

1. Bell Shaped

The distribution has a bell shaped curve.

2. Symmetric

The left and right sides are the mirror images.

Therefore:

$$P(X < \mu) = P(X > \mu)$$

3. Mean = Median = Mode

For a perfectly normal distribution:

Mean = Median = Mode

This happens because the distribution is perfectly symmetrical.

4. Total Area = 1

The total area under the normal curve represents total proabability.

$$\mathbf{P(-\infty < X < \infty) = 1}$$

or 100%

5. Tails Extend Infinitely

The curve approaches the x-axis but theoretically never actually reaches it.

Normal Distribution, Standard Normal Distribution its Properties

6. Defined by Mean and Standard Deviation

Mean(μ) Controls the center

Standard Deviation(σ) Control the spread

Effects of Mean and Standard Deviation

Example:

Suppose

$X \sim N(100,15)$

Meaning:

Mean = 100

Standard Deviation = 15

The center of the distribution is 100.

Now change the mean:

$X \sim N(120,15)$

The entire curve shifts to the right.

Changing the mean shifts the distribution.

Now Keep our mean constant:

$X \sim N(100,10)$

vs

$X \sim N(100,25)$

Changing the Standard Deviation changes the spread.

Emperical Rule: 68-95-99.7

μ = 100

For approximately normal data:

SD =10

Within 1 Standard Deviation

$\mu \pm 1\sigma$

Approximately:

68% of observations

Within 2 Standard Deviations

$\mu \pm 2\sigma$

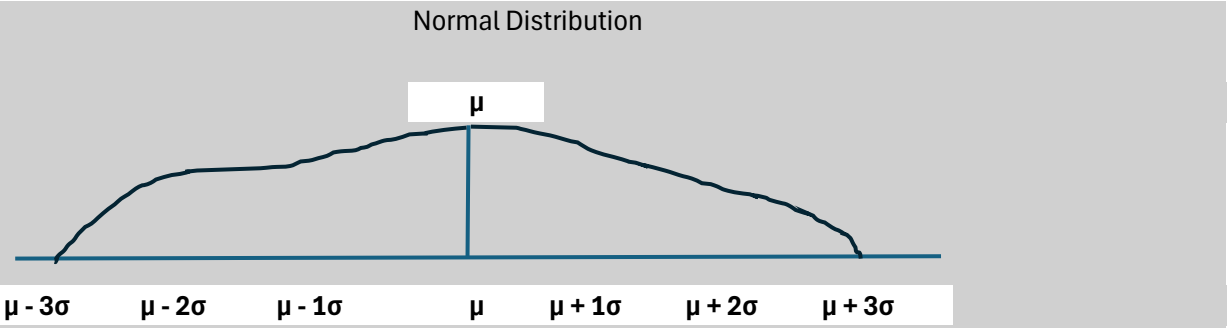
Approximately 95% of observations

Within Standard Deviations

$\mu \pm 3\sigma$

Approximately: 99.7%

Normal Distribution, Standard Normal Distribution its Properties



68% --> within μ - 1σ to μ + 1σ
95% --> within μ - 2σ to μ - 2σ
99.7% --> within μ - 3σ to μ - 3σ

Business Example:

Suppose delivery time is approximately normal:

μ = 30 Minutes
σ = 5 minutes

Approximately 68%

30-5 to 30+5

So :

25- 35 minutes

Approximately 95%

30-2(5) to 30+ 2(5)

So:

20- 40 minutes.

Approximately 99.7%

30- 3(5) to 30 + 3(5)

So:

15- 45 minutes

Business interpretation:

A company can use this to understand the expected operational range of delivery times.

Normal Distribution, Standard Normal Distribution its Properties

Standard Normal Distribution

The Standard normal Distribution has

$\mu = 0$

and

$\sigma = 1$

So every observation is expressed in terms of standard deviations from the mean.

Z-Score

A Z-Score tells us how many standard deviations an observation is away from the mean.

Formula: $z = (x- \mu)/\sigma$

x = observed values

μ = Mean

σ = Standard Deviation

Example:

Suppose:

Average Daily time (mu) = 30

Standard Deviation (SD) = 5

A particular delivery took (x) = 40

z = (40-30)/5 2

This delivery time is 2 Standard deviation above the average.

SD = 5

10 above the average.

Negative Z-Score

Suppose:

x = 20

z =(20-30)/5 -2

This delivery time is 2 Standard Deviations below the average.

Z-Score Business Applications

1. Delivery Analytics:

Compare delivery times across different cities.

Normal Distribution, Standard Normal Distribution its Properties

2. Employee Performance
Compare an employee's performance against the department average.
3. Credit Risk
Identify observations that are usually high or low compared with a reference distribution.
4. Customer Spending
Identify the customers whose spending is significantly above or below the average.
5. Manufacturing
Detect unsusual measurements or potential process anomalies.

Important Classification
A high positive Z-Score does not automatically means "good"

It simply means:
The value is above the mean by that many standard deviations.

For Example:
High Sales: --> Potentially good
High Delivery time:--> Potentially bad
High defect count --> bad

Delivery time(X)	40
Mean	30
SD	5
	2

STANDARDIZE Function is used to calculate the Z score in Excel.

Why?
We are standardizing the observations so that we can compare it with observations from another distribution.

Normal Distribution, Standard Normal Distribution its Properties

Introduction to Sampling Distribution

Suppose a company has:
100,000 customers
We don't want to survey all these 100,000 cutomers every time.
Instead we take a sample.

Example:

n = 100 customers

Calculate the average spending:

Suppose:

Xbar = ₹2,450

Taking another sample of 100

Xbar = ₹2,510

Another sample:

Xbar = ₹2,420

Another:

Xbar = ₹2,480

The sample means are not exactly same.

This variation creates a new distribution

SAMPLING DISTRIBUTION

A sampling Distribution is the distribution of a statistic obtained from **many possible samples of the** same size from a population.

Technically:
The Sampling Distribution of a statistc describes the probability distibution of that statistic across repeated random samples of a specified size of population.

Population Distribution:
Distribution of individual values in the population
Sample:
One subset selected from the population.
Sampling Distribution:
Distribution of a statistic -- such as the sample mean--across many possible samples.

Example: 1 million cutomers
The company wants to estimate: **Average monthly customer spending.**
Instead of analyzing each and every customer we cam take a sample of 100 customers.
We calculate our Xbar(Mean) and repeat this process thousands of times.

Normal Distribution, Standard Normal Distribution its Properties

Q1. What are three measures of central tendency in a perfect normal distribution?
Q2. What is the total area under the curve?
Q3. What does the standard deviation controls in a normal distribution?
Q4. What is 68-95-99.7 rule?
What is Standard normal Distribution?
mu = 0, sigma = 1